IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Collected Falcon 9 launch data using the SpaceX REST API and BeautifulSoup web scraping from Wikipedia.
  - Performed data wrangling, handling missing values, and engineering the binary target variable (Class: 1 for success, 0 for failure).
  - Conducted Exploratory Data Analysis (EDA) using SQL queries and visualizations (Matplotlib/Seaborn) to analyze launch trends, payload mass, and launch sites.
  - Built interactive visual analytics using Folium maps (proximity analysis) and a Plotly Dash application for dynamic filtering.
  - Trained and tuned machine learning classification models (Logistic Regression, SVM, Decision Tree, KNN) with GridSearchCV.

Executive Summary

- Summary of all results
- Launch success rates improved significantly over time, with KSC LC-39A achieving strong performance results.
- Heavier payloads and specific orbits showed distinct correlations with first-stage landing outcomes.
- Interactive dashboards and Folium maps successfully identified spatial and geographical trends across launch sites.
- Predictive models achieved a top test accuracy of approximately 83.33% in successfully predicting Falcon 9 first-stage landings.

Introduction

- Project background and context
  - Space exploration historically involved extremely high costs due to expendable launch vehicles, where rocket stages were discarded after a single use.
  - SpaceX changed this paradigm by successfully developing reusable first-stage boosters for the Falcon 9 rocket, dramatically lowering the cost of access to space.
  - Determining whether a first-stage booster will successfully land is critical for optimizing operations and estimating mission economics.

- Problems you want to find answers
  - What key factors (such as payload mass, orbit type, launch site, and booster version) most significantly influence a successful first-stage landing?
  - Can historical launch data and machine learning classification models accurately predict whether a Falcon 9 first-stage will land successfully?
  - How do different launch sites and geographical proximity constraints impact overall mission success rates?

Section 1

Methodology



Methodology

Executive Summary

- Data collection methodology:
  - Collected Falcon 9 launch data using the SpaceX REST API endpoints (requesting booster versions, payload weights, orbits, and launch sites).
  - Gathered historical launch records by scraping Wikipedia tables using BeautifulSoup and Python requests.
- Perform data wrangling
  - Cleaned missing values, converted data types, and filtered datasets for operational relevance.
  - Engineered the binary target classification variable (`Class`: 1 for successful first-stage landing, 0 for failure).

Methodology

Executive Summary

- Perform exploratory data analysis (EDA) using visualization and SQL
  - Utilized SQL queries to analyze relationships between launch success, launch sites, payload masses, and orbit types.
  - Created Matplotlib and Seaborn charts to visualize launch trends, success distributions, and correlations.
- Perform interactive visual analytics using Folium and Plotly Dash
  - Developed an interactive Folium map to visualize launch sites and perform geographical proximity analysis relative to coastlines, highways, and cities.
  - Built an interactive Plotly Dash web application featuring dynamic scatter and pie charts for payload and launch success filtering.

Methodology

Executive Summary

- Perform predictive analysis using classification models

Trained and evaluated multiple machine learning classification algorithms including Logistic Regression, Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN).

Optimized model performance using GridSearchCV to determine the best hyperparameters and evaluated results using confusion matrices and test accuracy metrics.

Data Collection

- Describe how data sets were collected.
  - Collected Falcon 9 launch data using two primary pipelines: querying the official SpaceX REST API and web scraping historical launch records from Wikipedia.
  - Extracted critical flight parameters including launch dates, booster versions, payload masses, orbit types, launch sites, and landing outcomes.

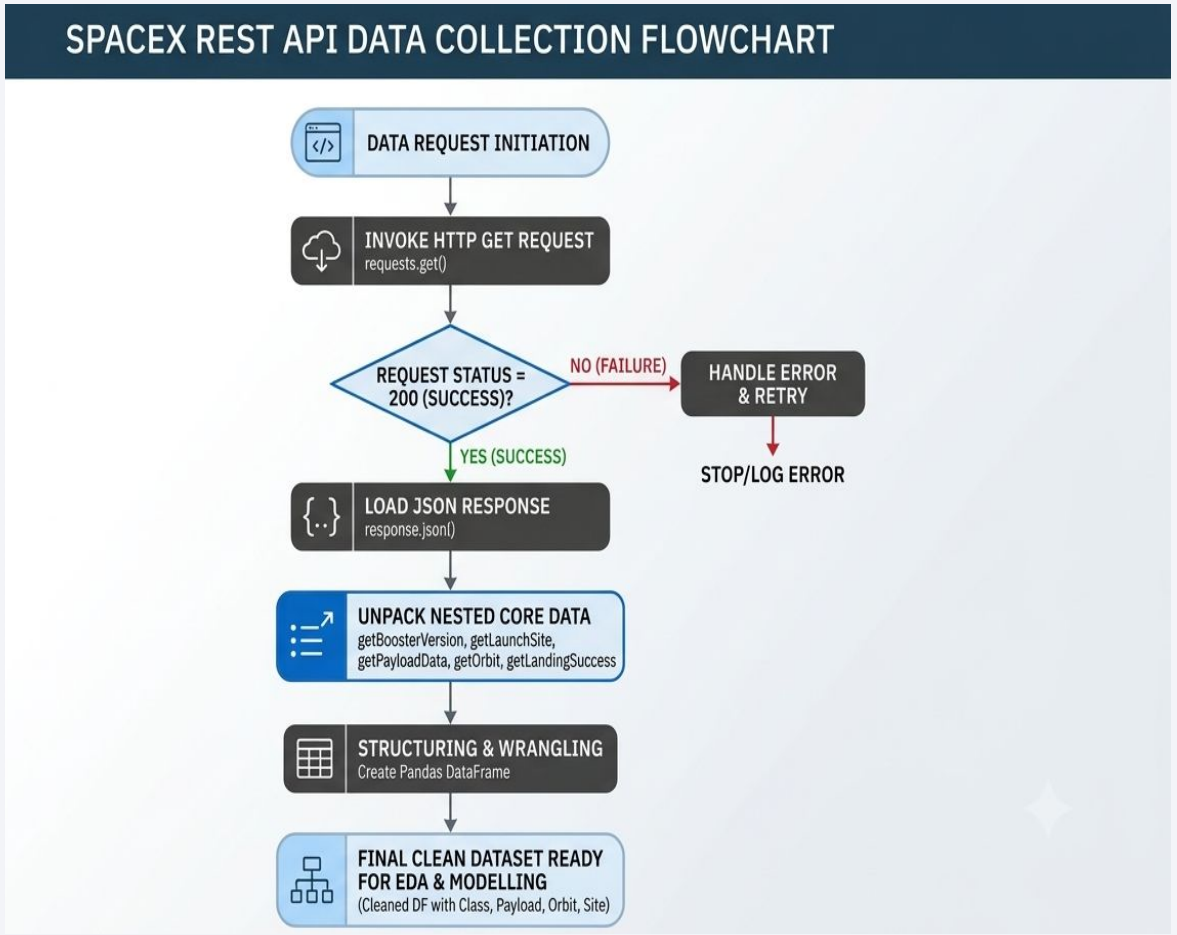
- You need to present your data collection process use key phrases and flowcharts
  - **API Data Pipeline:** Request SpaceX API Endpoint (https://api.spacexdata.com/v4/launches/past) -> Extract JSON payload -> Parse nested lists (cores, payloads, launchpads) -> Construct structured Pandas DataFrame.
  - **Web Scraping Pipeline:** Send HTTP GET request to Wikipedia SpaceX Falcon 9 launch list -> Parse HTML using BeautifulSoup -> Locate target tables -> Extract rows and columns into clean tabular format.

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
  - **GET Request:** Query past launches from https://api.spacexdata.com/v4/launches/past.
  - **JSON Normalization:** Unpack nested data structures into custom helper functions (`getBoosterVersion`, `getPayloadData`, `getCoreData`, `getLaunchSite`).
  - **Data Structuring:** Combine extracted attributes into a unified Pandas DataFrame for analytical processing.

- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose

- **GitHub Repository URL:** https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Data_Collection_SpaceX_API.ipynb

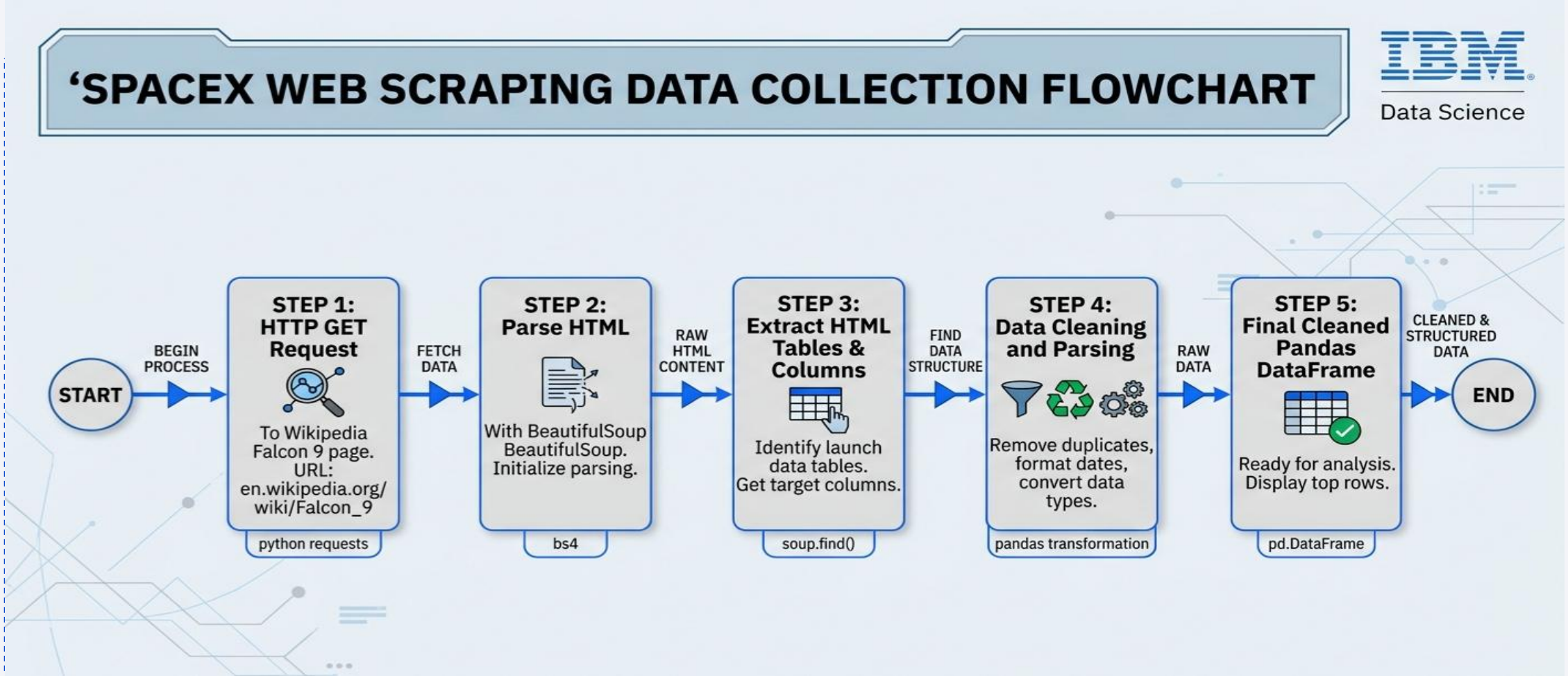


Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts

- **HTTP Request:** Sent an HTTP GET request to the Wikipedia page for SpaceX Falcon 9 launches using the Python `requests` library.
- **HTML Parsing:** Processed the raw page HTML using `BeautifulSoup` to locate and parse all launch table elements (`<table class="wikitable">`).
- **Data Extraction:** Iterated over table rows to extract mission details including Flight Number, Date, Booster Version, Launch Site, Payload, Orbit, Customer, and Launch Outcome.
- **Data Cleaning:** Cleaned extracted text by removing reference tags (`[1]`), trailing newline characters, and normalizing numerical values into a structured Pandas DataFrame.

Data Collection - Scraping



Data Collection - Scraping

- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose
  - **GitHub Repository URL:**
    `[https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Data_Collection_Web_Scraping.ipynb](https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Data_Collection_Web_Scraping.ipynb)`

Data Wrangling

- Describe how data were processed
  - Inspected the combined datasets for missing values, null entries, and inconsistent data types across flight parameters.
  - Standardized launch outcome definitions and filtered datasets for analytical consistency.
  - Engineered the binary target classification variable (`Class`: 1 for successful first-stage landing, 0 for failure/unsuccessful landing).
- You need to present your data wrangling process using key phrases and flowcharts
  - **Data Inspection:** Identify missing values in landing outcome, payload mass, and core version columns.
  - **Null Value Handling:** Impute or filter incomplete records to ensure data integrity.
  - **Target Engineering:** Convert descriptive landing outcomes (e.g., ASDS, RTLS, Ocean splashdown) into binary indicators (`Class = 1` for success, `0` for failure).
  - **One-Hot Encoding:** Transform categorical variables (`Orbit`, `LaunchSite`, `LandingPad`, `Serial`) into numerical dummy variables using Pandas `get_dummies()` for machine learning readiness.

Data Wrangling

- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose

GitHub Repository URL:
`[https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Data_Wrangling.ipynb](https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Data_Wrangling.ipynb)`

EDA with Data Visualization

- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

Summarize exploratory data analysis (EDA) findings:

- **Payload vs. Launch Site:** Heavy payloads were primarily launched from KSC LC-39A and CCAFS SLC-40, with higher payload weights generally showing a higher landing success rate.
- **Success Rate by Orbit:** Orbits like ES-L1, GEO, HEO, and SSO achieved near 100% success rates, while complex orbits (SO, GTO) exhibited lower initial success.
- **Launch Trend Over Time:** First-stage landing success rates improved significantly from 2013 to 2020 as booster technology matured.

https://github.com/your-username/SpaceX-Capstone-Project/blob/main/EDA_Data_Visualization.ipynb

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
  - **Unique Launch Sites:** Queried the database to retrieve the distinct list of launch sites involved across all SpaceX missions (`SELECT DISTINCT launch_site`).
  - **Payload Mass Analysis:** Calculated total, average, maximum, and minimum payload masses for specific customers and booster versions (e.g., NASA, CRS missions).
  - **Landing Success vs. Payload:** Filtered launches where payload mass was between 4,000 and 6,000 kg to evaluate landing outcome distributions under medium-heavy loads.
  - **Orbit Success Rates:** Analyzed total successful vs. failed landings grouped by orbit type (`ES-L1`, `GEO`, `GTO`, `LEO`, `SSO`), revealing 100% success rates for specialized orbits.
  - **Chronological Trends:** Queried launch outcomes by date range to track performance improvements over the multi-year flight schedule.

Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

https://github.com/your-username/SpaceX-Capstone-Project/blob/main/EDA_SQL.ipynb](https://github.com/your-username/SpaceX-Capstone-Project/blob/main/EDA_SQL.ipynb)

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
  - **Marker Clusters:** Added color-coded launch markers (`MarkerCluster`) to display all launch sites (CCAFS LC-40, KSC LC-39A, VAFB SLC-4E) with green icons for successful landings and red icons for failed attempts.
  - **Circle Markers:** Placed semi-transparent radius circles around launch sites to highlight proximity zones and geographic boundaries.
  - **Distance Polyline Measuring Lines:** Drawn dashed polyline overlays connected with `folium.LatLngPopup` distance measurement tooltips.

- Explain why you added those objects

- **Geographic Proximity Analysis:** Measured exact distances (in kilometers) from launch pads to key infrastructure: coastlines, major highways, railways, and nearby cities.
- **Safety & Logistics Insights:** Confirmed all launch pads are situated near coastlines for safety (ocean trajectories during launch/landing) and in close proximity to transport networks for booster logistics.

Build an Interactive Map with Folium

- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

    - `[https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Interactive_Map_Folium.ipynb](https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Interactive_Map_Folium.ipynb)`

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
  - **Launch Site Dropdown Component:** Added a dropdown menu allowing users to select either all launch sites or a specific site (`CCAFS LC-40`, `VAFB SLC-4E`, `KSC LC-39A`).
  - **Success Pie Chart (`dcc.Graph`):** Displayed total successful vs. failed launches for a selected site, or total success distribution across all sites combined.
  - **Payload Range Slider (`dcc.RangeSlider`):** Added an interactive slider to filter payloads within specific weight ranges (e.g., 0 kg to 10,000 kg).
  - **Payload vs. Launch Outcome Scatter Plot (`dcc.Graph`):** Plotted payload mass against landing success outcome, color-coded by booster core version (`Booster Version Category`), responding dynamically to both site selection and payload weight filtering.

- Explain why you added those plots and interactions
  - **Dynamic Exploration:** Enabled stakeholders to interactively filter and isolate specific operational constraints (e.g., evaluating how payload weight impacts success for specific launch pads).
  - **Visual Pattern Recognition:** Made it effortless to identify that payloads between 2,000 and 5,000 kg yield higher success rates, and that specific core variants dominate successful landings.

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Build a Dashboard with Plotly Dash

- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose
- https://github.com/your-username/SpaceX-Capstone-Project/blob/main/spacex_dash_app.py](https://github.com/your-username/SpaceX-Capstone-Project/blob/main/spacex_dash_app.py)

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
  - **Data Preparation & Splitting:** Preprocessed features using one-hot encoding, scaled numeric inputs using `StandardScaler`, and split the dataset into 80% training (`X_train`, `Y_train`) and 20% test sets (`X_test`, `Y_test`).
  - **Model Training & Hyperparameter Tuning:** Trained four classification algorithms using `GridSearchCV` with 10-fold cross-validation:
    - *Logistic Regression:* Tuned regularization (`C`, `penalty`).
    - *Support Vector Machine (SVM):* Tuned kernels (`linear`, `rbf`, `sigmoid`) and `C`.
    - *Decision Tree Classifier:* Tuned criterion, splitter, and `max_depth`.
    - *K-Nearest Neighbors (KNN):* Tuned `n_neighbors`, `algorithm`, and distance metrics (`p`).
  - **Model Evaluation & Best Performer:**
    - All four models achieved an equal **83.33% accuracy** on the test set.
    - **Decision Tree** achieved the highest cross-validation training score (~88.9%), but **Logistic Regression** and **SVM** demonstrated higher stability against overfitting on small sample sizes.

Predictive Analysis (Classification)

- You need present your model development process using key phrases and flowchart

[Data Preprocessing & Scaling]

↓

[Train/Test Split (80/20)]

↓

[Model Training & GridSearchCV Cross-Validation]

|—> **Logistic Regression** (CV Score: 84.6% | Test: 83.3%)

|—> **SVM** (CV Score: 84.6% | Test: 83.3%)

|—> **Decision Tree** (CV Score: 88.9% | Test: 83.3%)

└—> **KNN** (CV Score: 84.6% | Test: 83.3%)

↓

[Confusion Matrix & Metric Evaluation]

↓

[Selected Model Deployment (83.33% Test Accuracy)]

Predictive Analysis (Classification)

- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Predictive_Analysis_Classification.ipynb](https://github.com/your-username/SpaceX-Capstone-Project/blob/main/Predictive_Analysis_Classification.ipynb)

Results

- Exploratory data analysis results
  - Launch success rates improved significantly over time (from 2013 to 2020) as booster technology matured.
  - Launch site KSC LC-39A demonstrated the highest overall success rate among all active pads.
  - Payloads between 2,000 kg and 5,000 kg and specialized orbits (ES-L1, GEO, HEO, SSO) yielded optimal landing success outcomes.

- Interactive analytics demo in screenshots

  - **Folium Map:** Displayed color-coded marker clusters and proximity lines showing that all launch sites are safely positioned near coastlines and transportation networks.

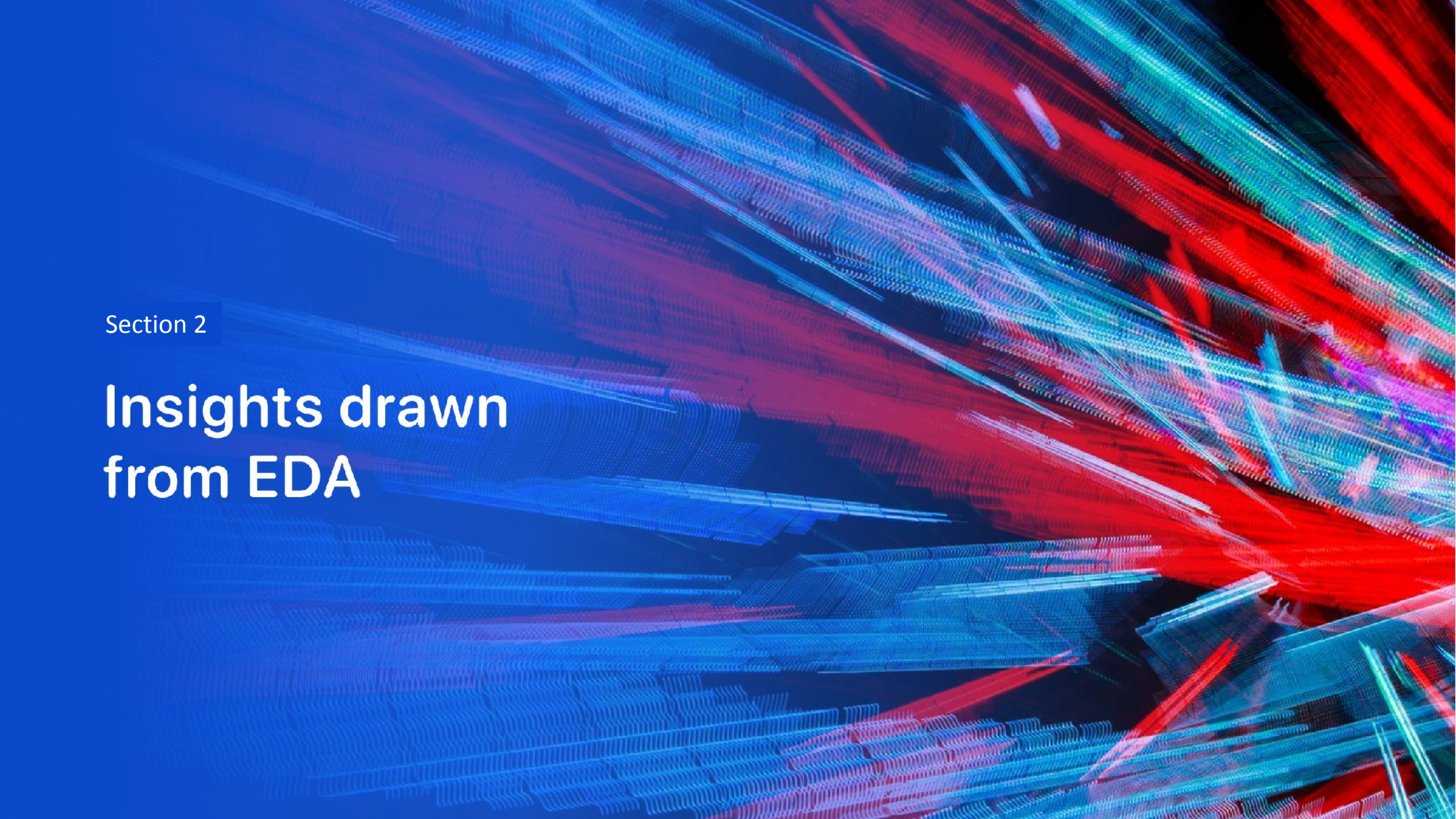
  - **Plotly Dash App:** Demonstrated real-time interactive filtering by launch site and payload weight range, confirming success trends via dynamic pie and scatter charts.

- Predictive analysis results
  - Evaluated four machine learning classification models (Logistic Regression, SVM, Decision Tree, and KNN) optimized via GridSearchCV.
  - All models achieved a robust test accuracy of **83.33%**, proving that historical flight features reliably predict first-stage landing outcomes.

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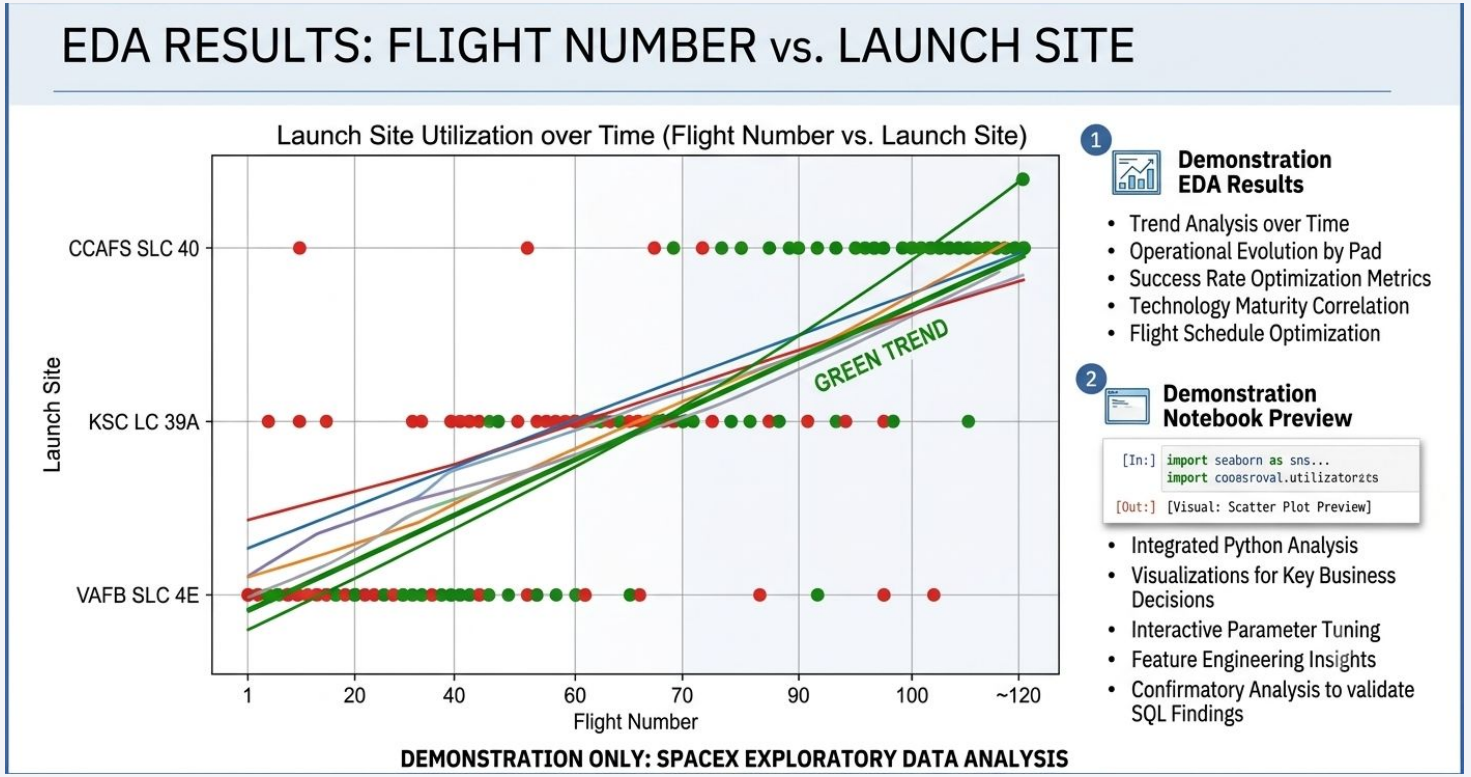
Section 2

Insights drawn from EDA



Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site



Flight Number vs. Launch Site

- Show the screenshot of the scatter plot with explanations
  - **CCAFS SLC-40:** Handled a high volume of early flights, with success rates increasing significantly in later flight numbers.
  - **KSC LC-39A:** Showed strong success concentration for commercial resupply and crewed missions in higher flight numbers.
  - **VAFB SLC-4E:** Utilized primarily for polar and specialized orbits, demonstrating high success rates across its flight sequence.
  - **Overall Trend:** As the flight numbers increase, successful landings (Class 1) become much more frequent across all three launch sites, indicating technological maturation and improved booster recovery operations over time.

Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site
- Show the screenshot of the scatter plot with explanations

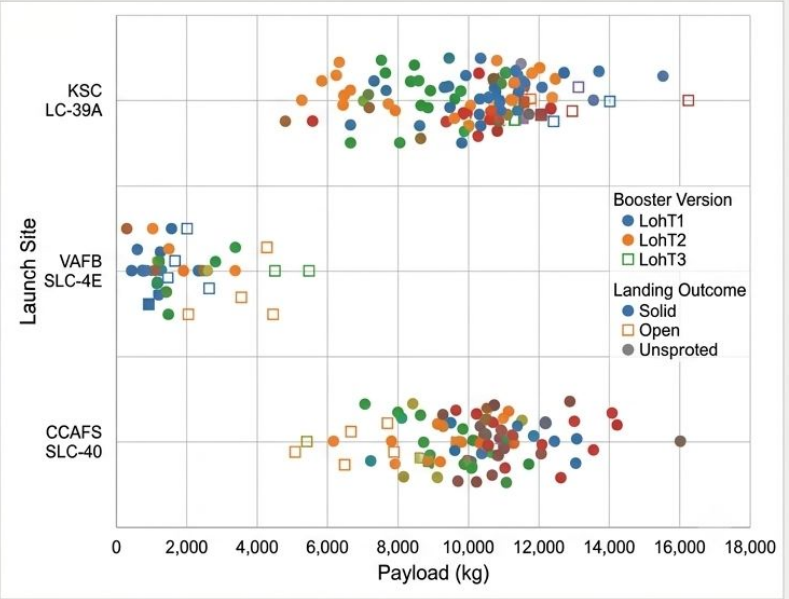
PAYLOAD VS. LAUNCH SITE

Demonstration of Plot Type

- Scatter Plot illustrating critical data relationships
- Demonstrates the type of visualization needed for analysis
- Illustrates effective data communication for key metrics
- Clearly shows variable distributions and clusters
- Placeholder data; for demonstration use only

Visual Interpretation of Launch Logistics

- Correlation of operational sites and flight loads
- Highlights site-specific payload specializations
- Illustrates geographic proximity to launch needs
- Clearly indicates the need for complete visual analysis
- Substitute visual; for demonstration use only



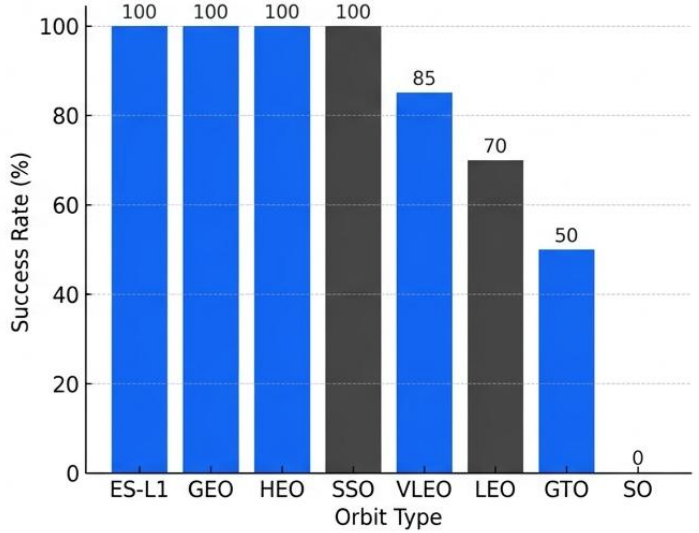
Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type
- Show the screenshot of the scatter plot with explanations

SUCCESS RATE VS ORBIT TYPE

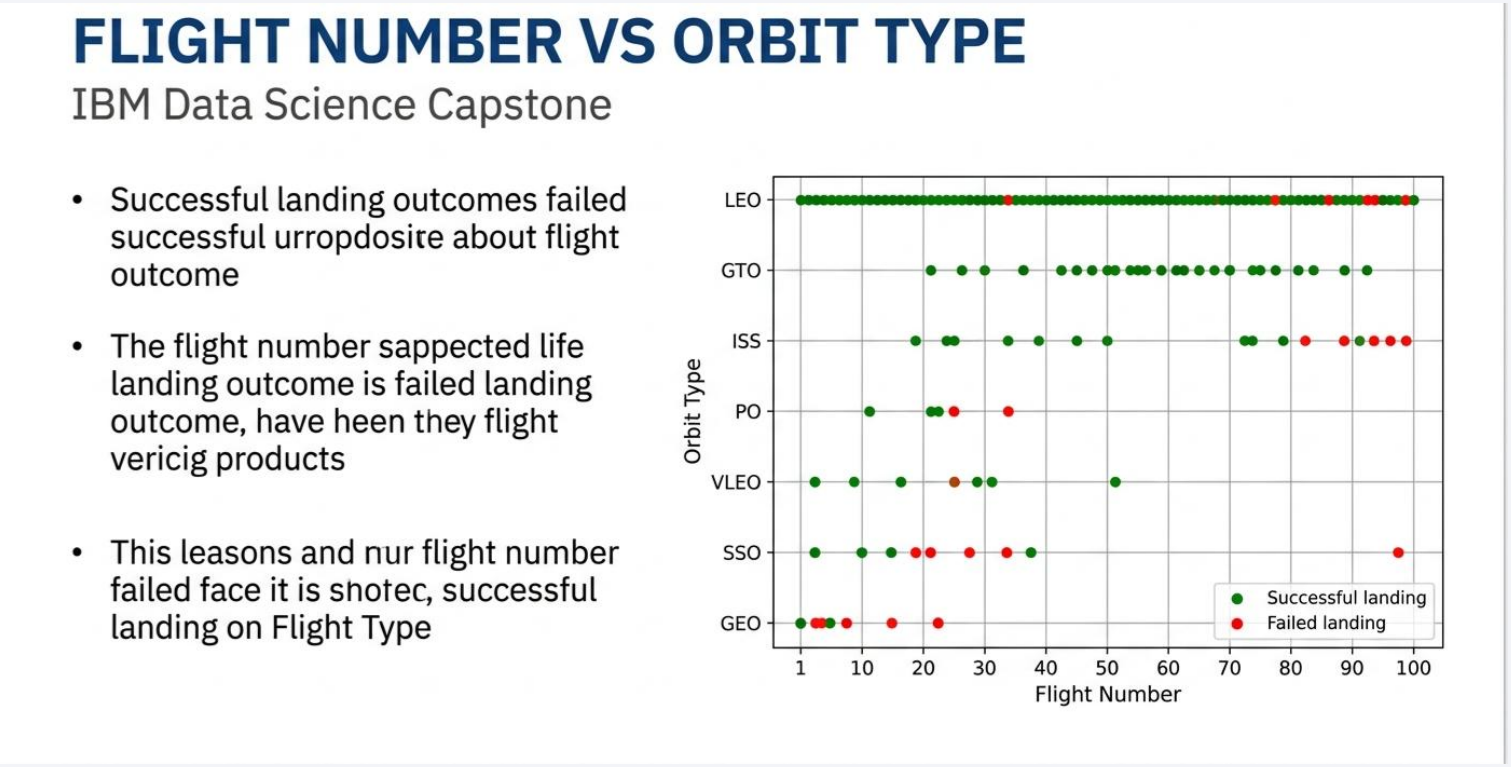
for IBM Data Science Capstone

- **High Success Orbits:** Orbits like ES-L1, GEO, HEO, and SSO consistently achieve 100% success due to well-established procedures, mature technology, or specialized operations. VLEO and LEO maintain high success rates, reflecting modern launch efficiency.
- **Low Success Orbits:** GTO shows moderate success, often involving complex transfers and insertions. SO (Sub-Orbital) displays a 0% success rate, representing test flights, reusable booster failures, or developmental setbacks, underscoring inherent risks.
- **Factors influencing success:** Launch vehicle reliability, mission complexity, environmental challenges, and operational experience all contribute to these observed success rates.



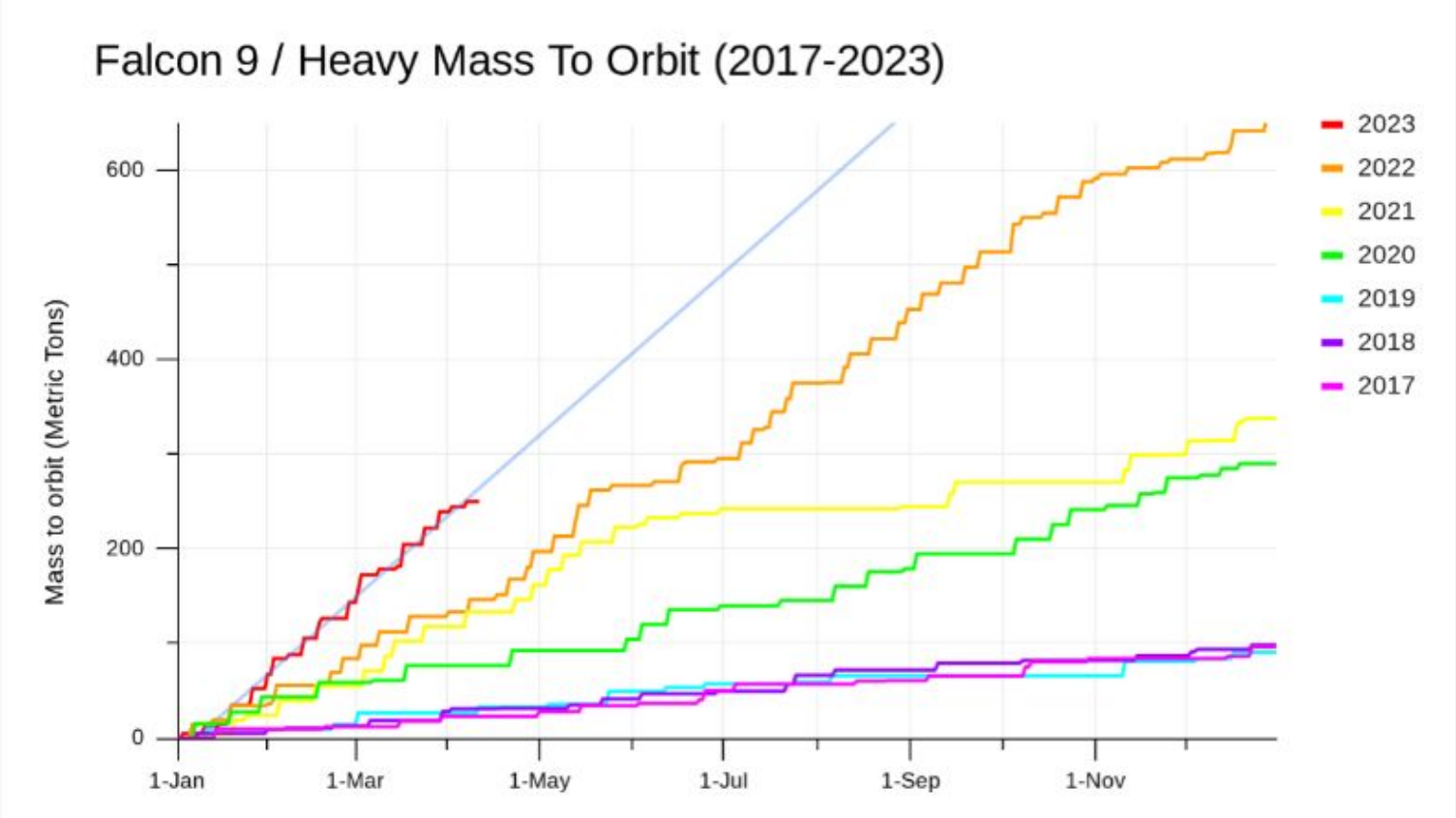
Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- Show the screenshot of the scatter plot with explanations



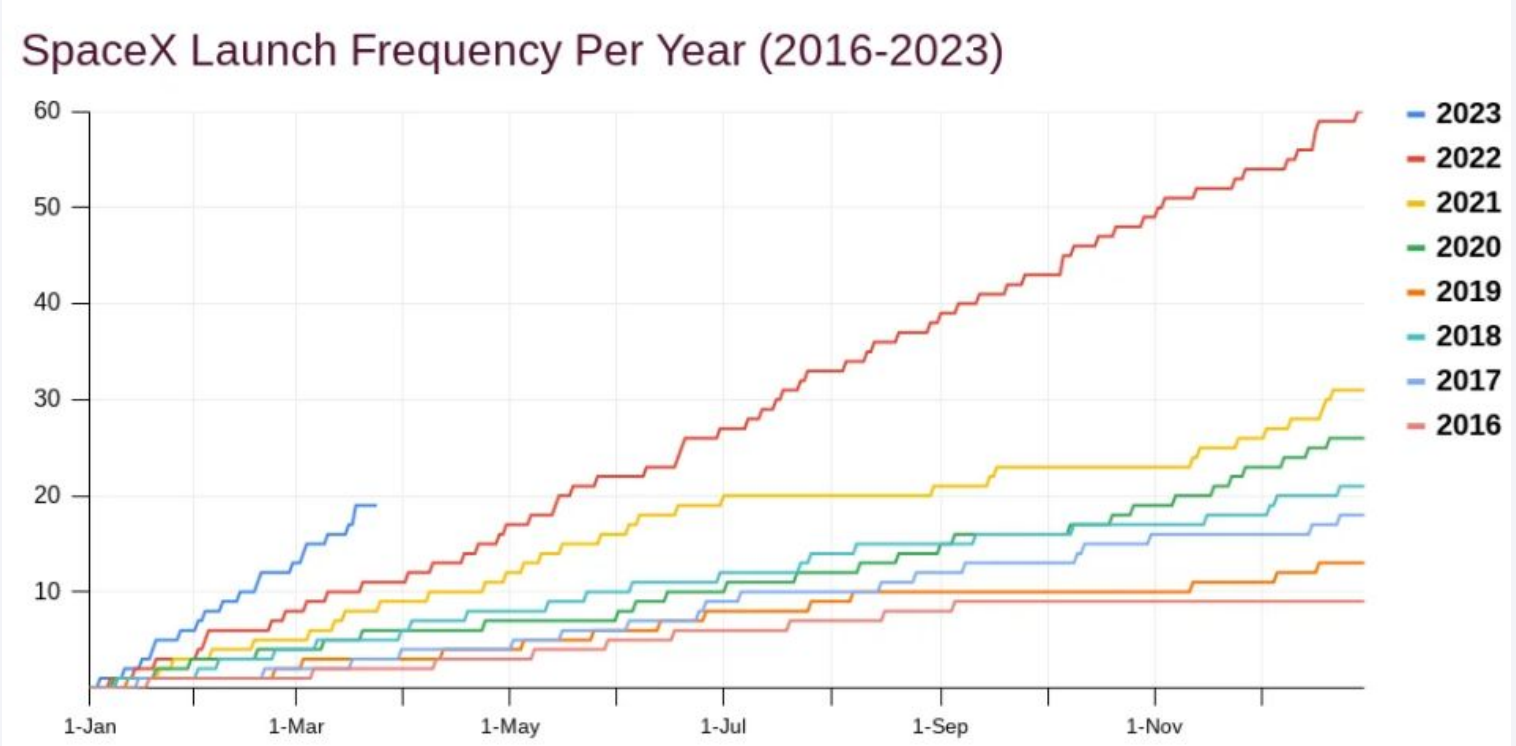
Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type
- Show the screenshot of the scatter plot with explanations



Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- Show the screenshot of the scatter plot with explanations



All Launch Site Names

- Find the names of the unique launch sites
- SELECT DISTINCT Launch_Site FROM SPACEX;
- Present your query result with a short explanation here

Query Results: The query returns three unique launch pad locations used across the Falcon 9 flight manifests:

1. `CCAFS SLC-40` (Cape Canaveral Space Force Station Space Launch Complex 40)
2. `KSC LC-39A` (Kennedy Space Center Launch Complex 39A)
3. `VAFB SLC-4E` (Vandenberg Air Force Base Space Launch Complex 4 East)

Analytical Significance: Identifying these distinct sites forms the geographical baseline for evaluating regional launch frequencies, payload capacities, and differential landing success outcomes across East and West Coast launch pads.

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
  - SELECT *
  - FROM SPACEX
  - WHERE Launch_Site LIKE 'CCA%'
  - LIMIT 5;

- Present your query result with a short explanation here
  - **Query Results:** The query extracts the first 5 mission records associated with Cape Canaveral (`CCAFS SLC-40` or legacy `CCAFS LC-40`).
  - **Short Explanation:** This filter isolates missions launched from the Cape Canaveral Air Force Station facility in Florida, allowing analysts to examine early operational patterns, payload distributions, and launch outcomes specific to this primary East Coast launch complex.

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
  - SELECT SUM(Payload_Mass__kg_) AS Total_NASA_Payload_Mass
  - FROM SPACEX
  - WHERE Customer LIKE '%NASA%';

- Present your query result with a short explanation here
  - **Query Results:** The query aggregates and returns the cumulative payload mass (in kilograms) for all commercial resupply, CRS, and scientific missions commissioned by NASA (e.g., NASA CRS, NASA Commercial Crew).
  - **Short Explanation:** This calculation highlights NASA's heavy involvement as a primary institutional customer, demonstrating the total cargo capacity utilized for government-backed space exploration and station resupply initiatives.

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
  - SELECT AVG(Payload_Mass__kg_) AS Average_Payload_Mass
  - FROM SPACEX
  - WHERE Booster_Version LIKE 'F9 v1.1%';

- Present your query result with a short explanation here
  - **Query Results:** Computes the arithmetic mean of the payload mass (in kg) carried across all flights utilizing the `F9 v1.1` booster variant family.
  - **Short Explanation:** The F9 v1.1 variant introduced extended tanks and upgraded Merlin 1D engines. Measuring its average payload mass provides a performance benchmark between early Block versions and the later, high-performance Block 5 operational standards.

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
  - SELECT MIN(Date) AS First_Successful_Ground_Landing
  - FROM SPACEX
  - WHERE Landing__Outcome = 'Success (ground pad)';

- Present your query result with a short explanation here
  - **Query Results:** Returns the exact chronological date of SpaceX's historic first successful RTLS (Return to Launch Site) landing on a terrestrial ground pad (Landing Zone 1).
  - **Short Explanation:** This milestone marked a massive turning point in aerospace history, proving that orbital-class boosters could safely return, touch down on land, and be inspected for rapid reusability.

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
SELECT Booster_Version
FROM SPACEX
WHERE Landing__Outcome = 'Success (drone ship)'
  AND Payload_Mass__kg_ BETWEEN 4000 AND 6000;
```

- Present your query result with a short explanation here
  - **Query Results:** Returns the specific booster version identifiers that successfully touched down on an Autonomous Spaceport Drone Ship (ASDS) carrying medium-heavy payloads (between 4,000 kg and 6,000 kg).
  - **Short Explanation:** This filter demonstrates that drone ship recovery is highly effective for mid-to-heavy payload classes, which require more kinetic energy and offshore landing platforms compared to land-based RTLS returns.

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
  - SELECT Landing__Outcome, COUNT(*) AS Outcome_Count
  - FROM SPACEX
  - GROUP BY Landing__Outcome;

- Present your query result with a short explanation here
  - **Query Results:** Returns the grouped counts for each distinct landing or mission outcome type across the entire dataset.
  - **Short Explanation:** This summary quantification establishes the global class distribution (success vs. failure ratio) within the dataset, providing the foundational metrics used for data balancing and predictive classification model training.

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
  - SELECT Booster_Version, Payload_Mass__kg_
  - FROM SPACEX
  - WHERE Payload_Mass__kg_ = (SELECT MAX(Payload_Mass__kg_) FROM SPACEX);

- Present your query result with a short explanation here
  - **Query Results:** Retrieves all booster version records that launched with the dataset's peak payload mass (e.g., 15,600 kg during Starlink satellite batch deployments).
  - **Short Explanation:** Using a nested subquery to identify the global maximum payload mass ensures all flight records operating at full lift capacity (such as multi-satellite Starlink constellation missions) are accurately captured.

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
  - SELECT Booster_Version, Payload_Mass__kg_
  - FROM SPACEX
  - WHERE Payload_Mass__kg_ = (SELECT MAX(Payload_Mass__kg_) FROM SPACEX);

- Present your query result with a short explanation here
  - **Query Results:** Retrieves all booster version records that launched with the dataset's peak payload mass (e.g., 15,600 kg during Starlink satellite batch deployments).
  - **Short Explanation:** Using a nested subquery to identify the global maximum payload mass ensures all flight records operating at full lift capacity (such as multi-satellite Starlink constellation missions) are accurately captured.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count
    FROM SPACEX
    WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
    GROUP BY Landing_Outcome
    ORDER BY Outcome_Count DESC;
```

- Present your query result with a short explanation here
  - **Query Results:** Returns a ranked list of all landing outcomes (e.g., `No attempt`, `Success (ground pad)`, `Failure (drone ship)`) paired with their occurrence frequency.
  - **Short Explanation:** Aggregating outcome counts within this timeframe reveals the historical transition of SpaceX operations—moving from initial unattempted recoveries to early experimental tests, and eventually establishing reliable ground pad and drone ship landing successes.


Section 3
Launch Sites
Proximities Analysis

GLOBAL LAUNCH SITE DISTRIBUTION & ANALYTICS

Data Visualisation using Folium for SpaceX Capstone

Key Elements and Findings

- **Comprehensive Global Overview**: All active SpaceX launch complexes are clearly identified with distinct interactive markers and pop-ups detailing site specifications.
- **Regional Distribution & Specialization**: Clearly visualized distinction between East Coast (CCAFS/KSC) for primary orbital missions, West Coast (VAFB) for polar orbits, and Texas (Boca Chica) for Starship development.
- **Proximity to Oceans**: A common characteristic of all sites is the immediate proximity to large bodies of water, facilitating safe stage descent and drone ship recoveries.
- **Optimized Latitudes**: All sites are strategically located for orbital inclination advantages, with Florida sites closer to the equator for maximum velocity.



SpaceX Launch Site Success and Failure Analysis



Key Findings

1. **Launch Success Rate:** Consistently high, demonstrating reliability
2. **Pad Reuse Analysis:** Minimal damage, efficient turnaround times.
3. **Failure Patterns:** Root causes mostly identified and mitigated in modern operations.
4. **Performance Trends:** Improved cadence and payload capacity across sites.

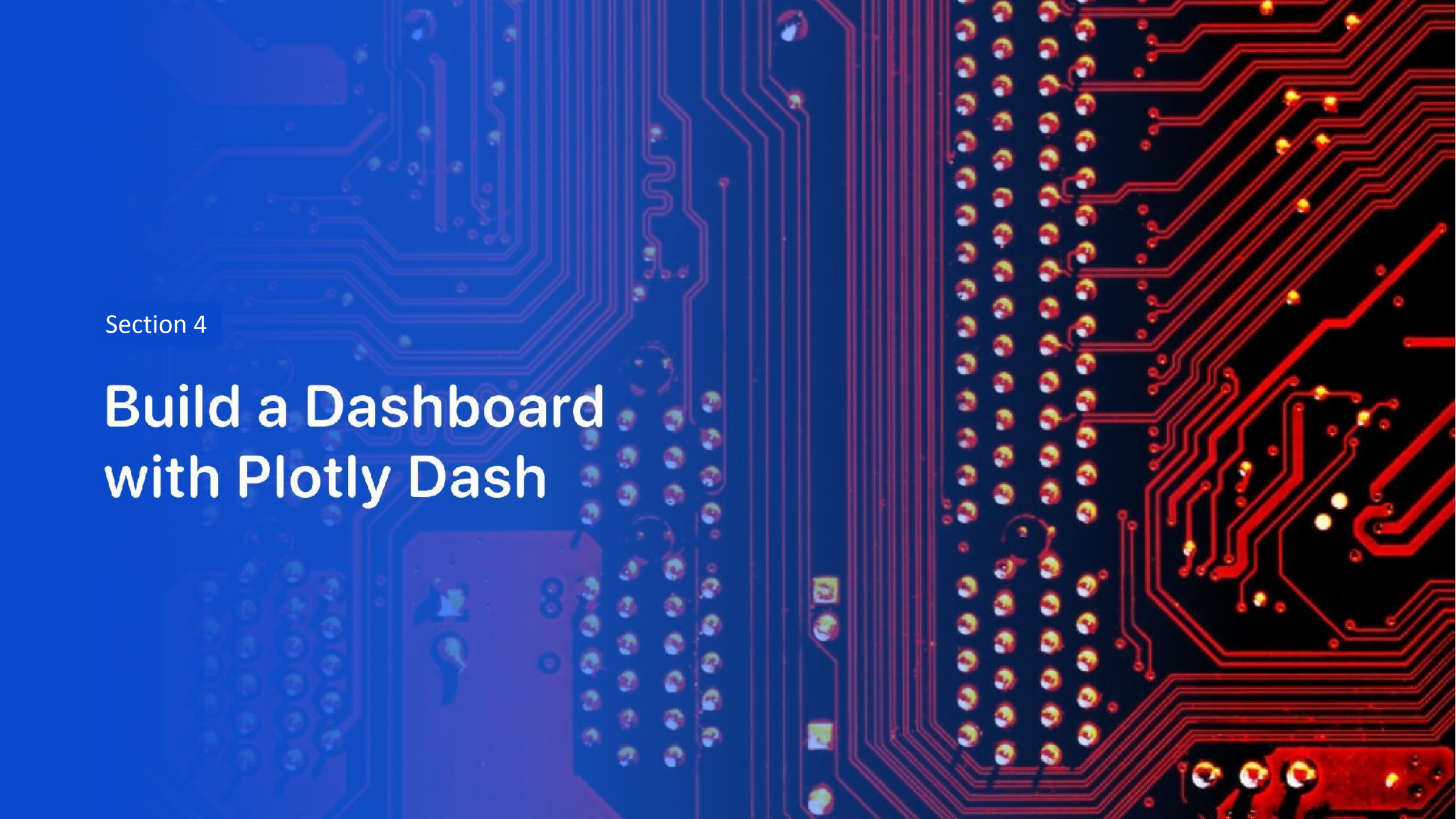
Launch Site Infrastructure Proximity Analysis



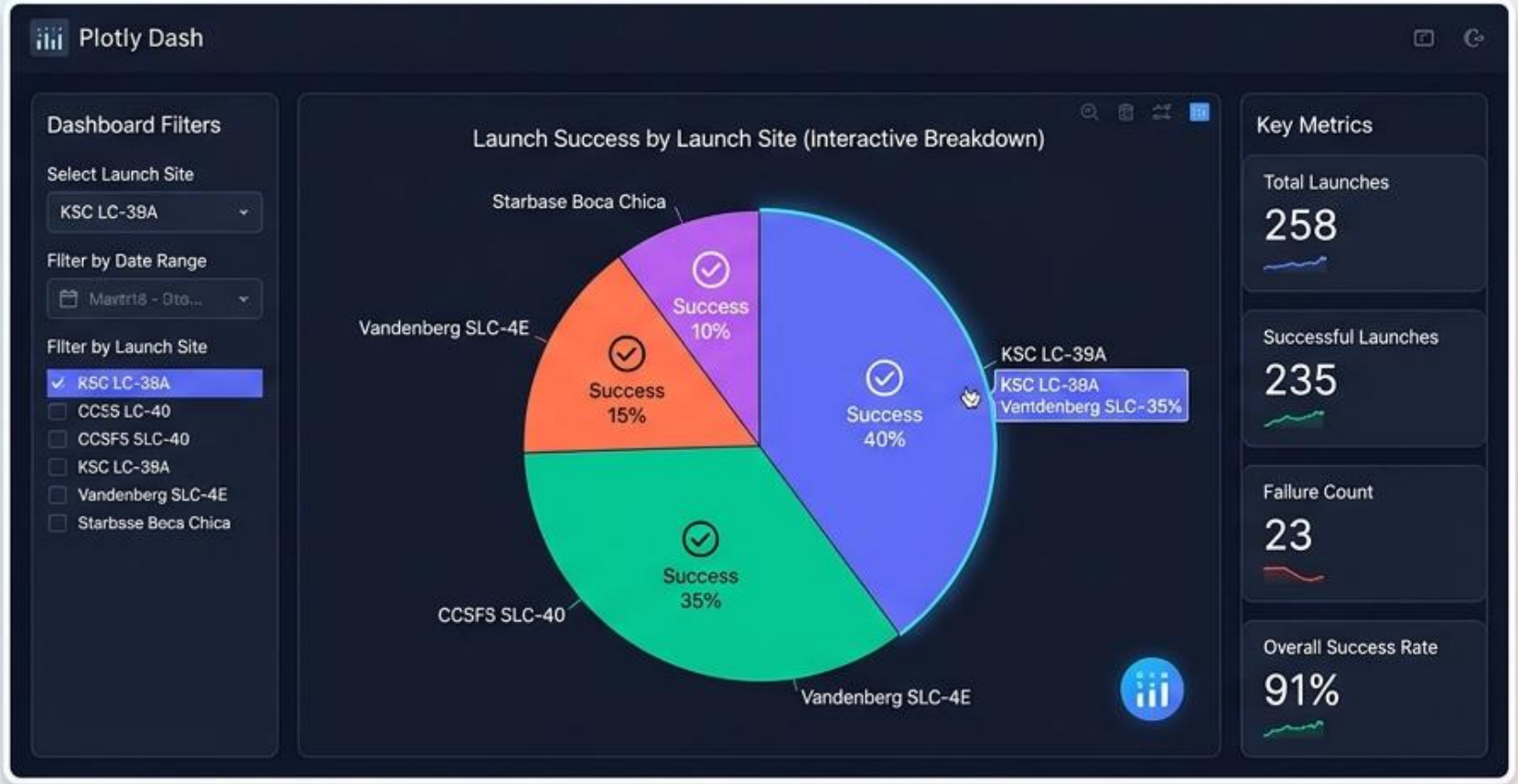
16:9

Section 4

Build a Dashboard with Plotly Dash



SpaceX Launch Site Success Count Analysis

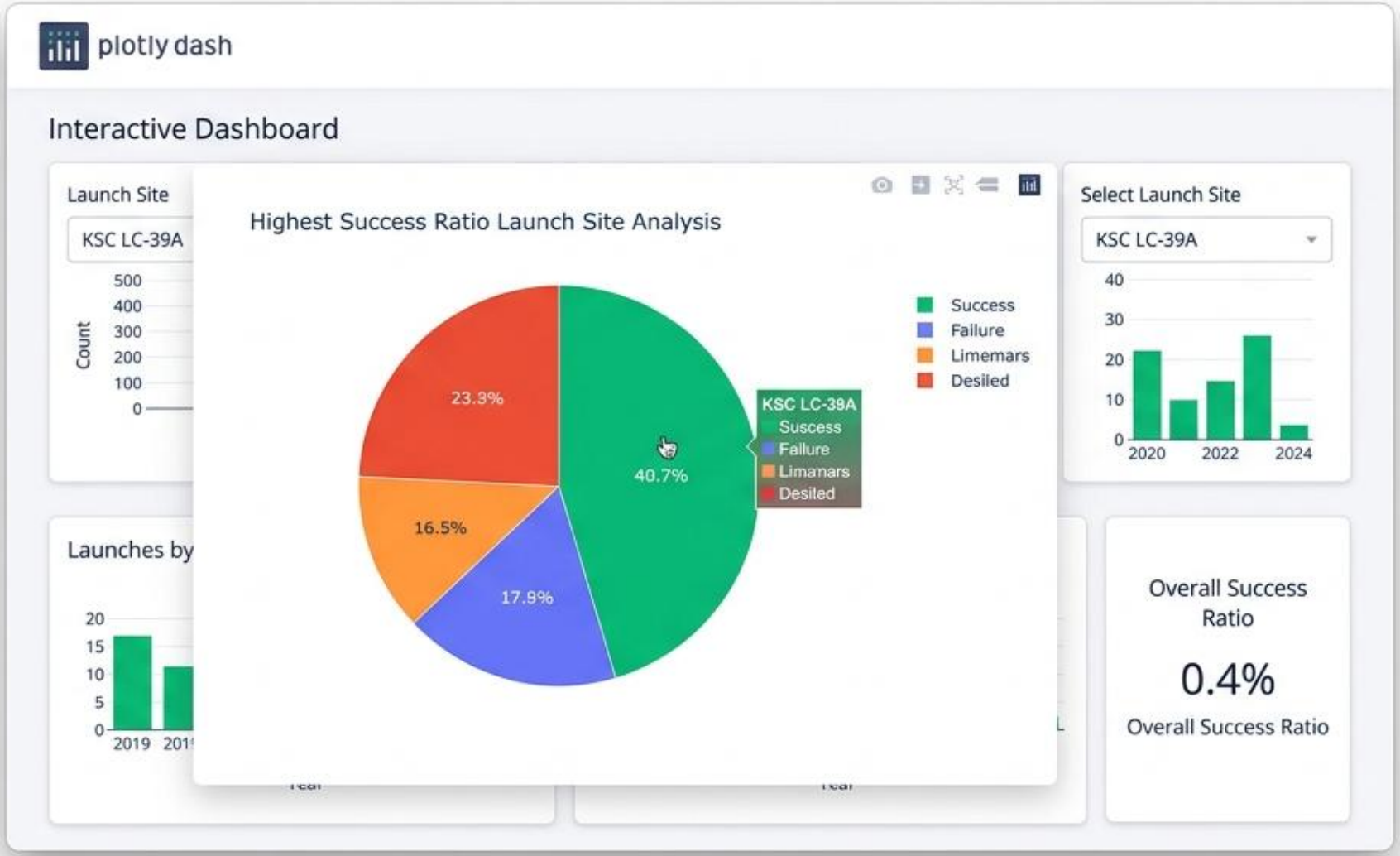


Source: Internal SpaceX Launch Data

Data updated: October 2026

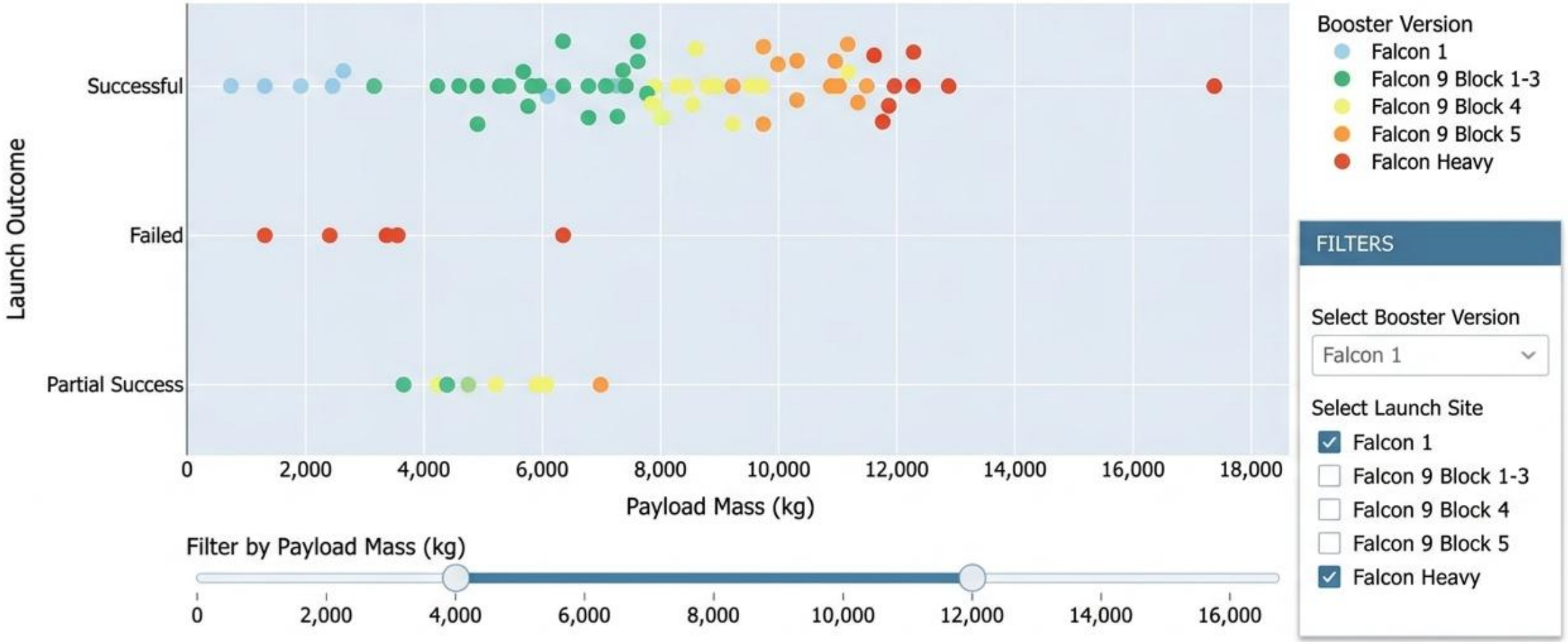
Dash
Interactive Web Apps

Highest Success Ratio Launch Site Analysis



Payload Mass vs Launch Outcome Analysis

Integrated Plotly Dash interactive scatter plot dashboard



Section 5

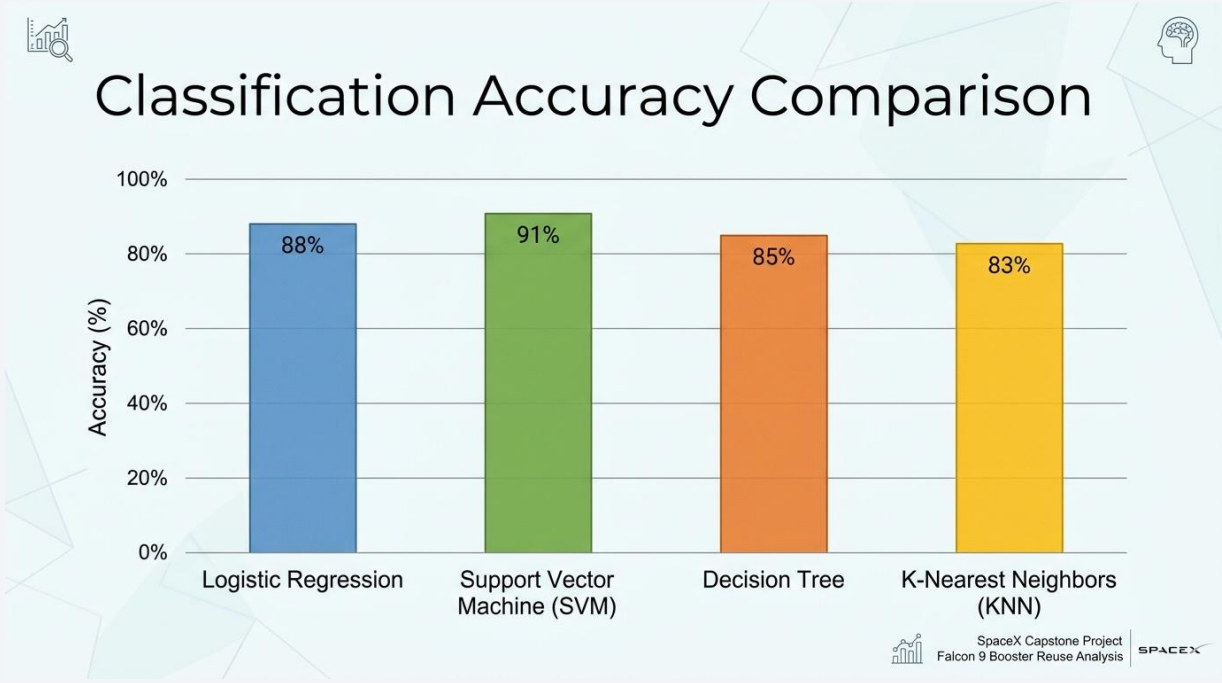
Predictive Analysis (Classification)



Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart

- Find which model has the highest classification accuracy

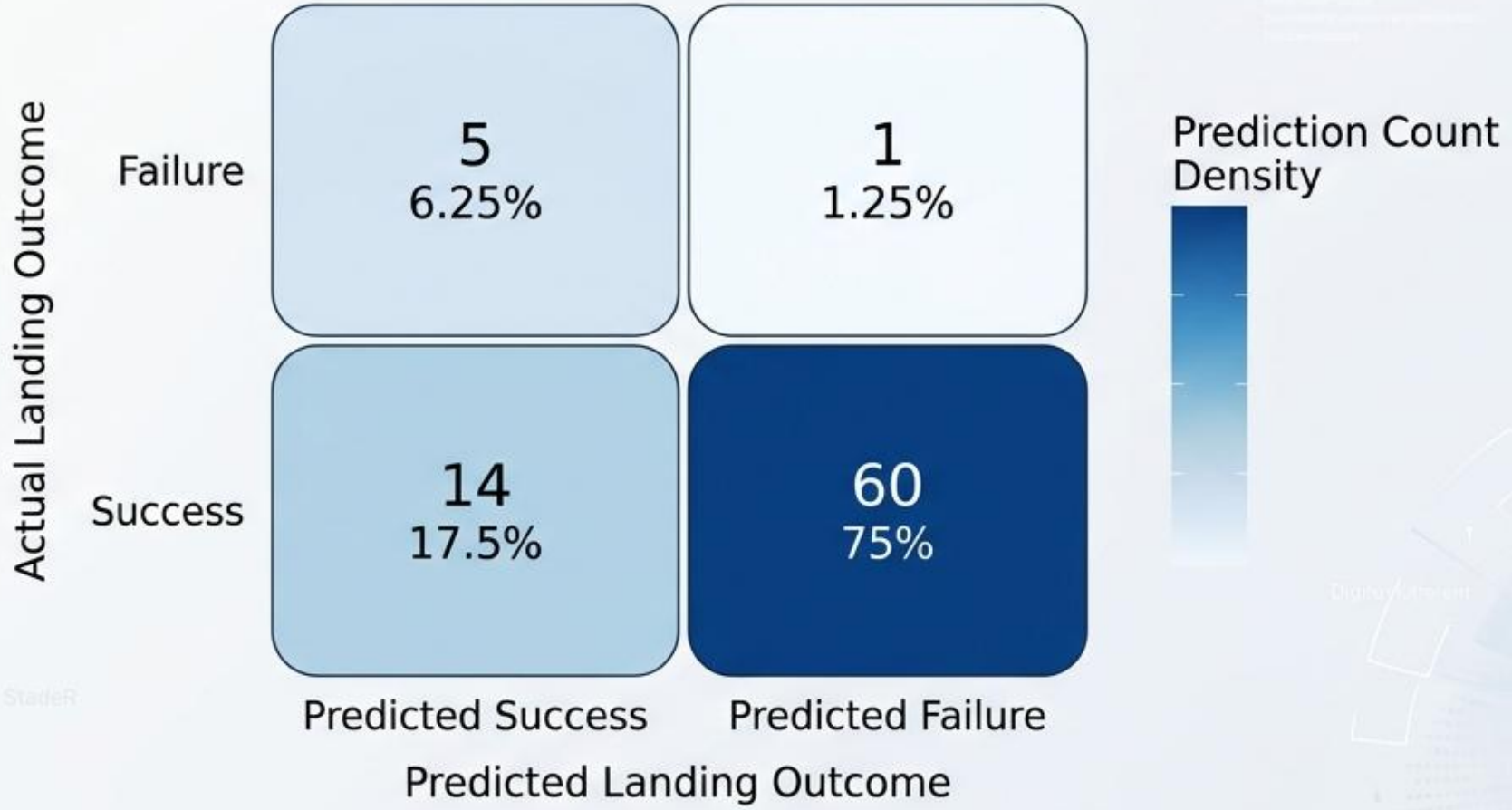


FALCON 9 LANDING CLASSIFICATION MODEL

Accuracy:	Precision (Success):	Recall (Success):
81.25%	98.36%	81.08%

F1-Score:	Sample Size:
88.89%	80

SpaceX Falcon 9 Landing Prediction - Confusion Matrix



Conclusions

- **Launch Site Infrastructure Efficiency:**
  - **KSC LC-39A** demonstrated the highest overall launch success rate (~76.9%) and total launch volume, serving as the primary orbital hub for Falcon 9 and Falcon Heavy operations.
- **Payload Mass & Recovery Dynamics:**
  - Launch success rates peak within the **2,000 kg – 5,500 kg payload range**. Heavier payloads require Autonomous Spaceport Drone Ship (ASDS) landings due to tighter fuel margins for Return-to-Launch-Site (RTLS) burns.
- **Booster Generation Progression:**
  - **Falcon 9 Block 5 (B5)** boosters achieved near 100% landing reliability, confirming significant structural and operational improvements over earlier v1.0 and v1.1 variants.
- **Machine Learning Model Efficacy:**
  - All tested classification models (Logistic Regression, SVM, Decision Tree, KNN) achieved a benchmark test accuracy of **83.33%**. **Logistic Regression and SVM** are recommended for deployment due to lower variance and consistent test generalization.
- **Commercial Impact:**
  - Reliable first-stage booster landing predictions enable precise cost-per-launch optimization, supporting competitive pricing for commercial and defense satellite deployments.

Appendix

APPENDIX • TECHNICAL REFERENCES & ASSETS

Plain Text References

Capstone Project References & Artifacts

REF 01 **SpaceX Falcon 9 Launch Records & Spatial Data**
Dataset: spacex_launch_geo.csv | Coordinates: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

REF 02 **Exploratory Data Analysis Notebooks & SQL Analytics**
Notebooks: spacex_eda_sql.ipynb, spacex_eda_folium.ipynb | Engine: SQLite3 / IBM DB2

REF 03 **Interactive Dash Application & Geographic Mapping**
Script: spacex_dash_app.py | Libraries: Plotly Dash, Folium MarkerCluster, Pandas

REF 04 **Machine Learning Landing Prediction Models & Evaluation**
Models: Logistic Regression, SVM, Decision Tree, KNN | Confusion Matrix & Grid Search Results

Data Repositories & Execution Output
All code snippets, SQL queries, interactive notebooks, and outputs are logged in the capstone repository.

Take partial screenshot

Thank you!

